

DataBase HW 1

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1. Answer Questions

- a. What the difference between type “char” and type “varchar”?

對於 char 來說，有沒有 var 的差別就在於有沒有定長，沒有 var 的 char 有固定長度，也就是輸入的英文長度未達固定長度時，會用空格來填充，有 var 的 varchar 再輸入長度未達時，並不會填充。因此 varchar 相較於 char 更省空間。

- b. Type “boolean” would be stored as which type in MySQL?

MySQL 中並沒有 boolean 這個型態，boolean 在裡面是作為 tinyint 的同義詞，0 視為 false，非 0 視為 true。

- c. How many bytes it should take for “tinyint”, “smallint”, “mediumint”, “int”? (e.g. 8 bytes for “bigint”) And what’s the range they can express? (e.g. from -1000 to 1000)

Tinyint: 1 byte, range: -128 ~ 127

Smallint: 2 byte, range: -32,768 ~ 32,767

Mediumint: 3 byte, range: -8,388,608 ~ 8,388,607

Int: 4 byte, range: -2,147,483,648 ~ 2,147,483,647

2. Tasks

A. Create Tables

```
mysql> describe customers;
```

Field	Type	Null	Key	Default	Extra
customer_id	varchar(50)	NO	PRI	NULL	
customer_unique_id	varchar(50)	YES		NULL	
customer_zip_code_prefix	int	YES		0	
customer_city	varchar(50)	YES		NULL	
customer_state	varchar(50)	YES		NULL	

```
5 rows in set (0.04 sec)
```

```
mysql> describe geolocation;
```

Field	Type	Null	Key	Default	Extra
geolocation_zip_code_prefix	varchar(50)	YES		NULL	
geolocation_lat	float	NO	PRI	NULL	
geolocation_lng	float	NO	PRI	NULL	
geolocation_city	varchar(50)	YES		NULL	
geolocation_state	varchar(50)	YES		NULL	

```
5 rows in set (0.00 sec)
```

```
mysql> describe order_items;
```

Field	Type	Null	Key	Default	Extra
order_id	varchar(50)	NO	PRI	NULL	
order_item_id	int	NO	PRI	0	
product_id	varchar(50)	YES		NULL	
seller_id	varchar(50)	YES		NULL	
shipping_limit_date	datetime	YES		NULL	
price	float	YES		NULL	
freight_value	float	YES		NULL	

```
7 rows in set (0.01 sec)
```

```
mysql> describe orders;
```

Field	Type	Null	Key	Default	Extra
order_id	varchar(50)	NO	PRI	NULL	
customer_id	varchar(50)	YES		NULL	
order_status	varchar(50)	YES		NULL	
order_purchase_timestamp	datetime	YES		NULL	
order_approved_at	datetime	YES		NULL	
order_delivered_carrier_date	datetime	YES		NULL	
order_delivered_customer_date	datetime	YES		NULL	
order_estimated_delivery_date	datetime	YES		NULL	

```
8 rows in set (0.00 sec)
```

```
mysql> describe payments;
```

Field	Type	Null	Key	Default	Extra
order_id	varchar(50)	NO	PRI	NULL	
payment_sequential	int	YES		0	
payment_type	varchar(50)	YES		NULL	
payment_installments	int	YES		0	
payment_value	float	YES		NULL	

```
5 rows in set (0.01 sec)
```

```
mysql> describe products;
```

Field	Type	Null	Key	Default	Extra
product_id	varchar(50)	NO	PRI	NULL	
product_category	varchar(50)	YES		NULL	
product_name_length	int	YES		0	
product_description_length	int	YES		0	
product_photos_qty	int	YES		0	
product_weight_g	int	YES		0	
product_length_cm	int	YES		0	
product_height_cm	int	YES		0	
product_width_cm	int	YES		0	

```
9 rows in set (0.00 sec)
```

```
mysql> describe sellers;
+-----+-----+-----+-----+-----+-----+
| Field                | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| seller_id            | varchar(50)   | NO   | PRI | NULL    |       |
| seller_zip_code_prefix | int           | YES  |     | 0        |       |
| seller_city          | varchar(50)   | YES  |     | NULL    |       |
| seller_state         | varchar(50)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.01 sec)
```

B. Query Tasks

1. Please list the number of all different clients. You must have **"COUNT"** syntax in usage of SQL.

```
mysql> source 1.sql
+-----+
| cnt  |
+-----+
| 96096 |
+-----+
1 row in set (0.32 sec)
```

2. Please list the number of different customer_unique_id which start with "ff9" . You must have **"DISTINCT"** syntax in usage of SQL.

```
mysql> source 2.sql
+-----+
| cnt |
+-----+
| 35  |
+-----+
1 row in set (0.03 sec)
```

3. Please list which 3 cities have the most orders made in state 'RJ' . You must sort counts in decreasing order and have **"ORDER BY"** syntax in usage of SQL.

```
mysql> source 3.sql
+-----+
| city          |
+-----+
| rio de janeiro |
| niteroi       |
| nova iguacu   |
+-----+
3 rows in set (0.32 sec)
```

4. Please list the top 3 order_id which cost the most days to deliver (the date of product arrival - the date the product starts to deliver). You must sort counts in decreasing order and have "ORDER BY" syntax in usage of SQL.

```
mysql> source 4.sql
+-----+
| order_id      |
+-----+
| 1b3190b2dfa9d789e1f14c05b647a14a |
| ca07593549f1816d26a572e06dc1eab6 |
| 285ab9426d6982034523a855f55a885e |
+-----+
3 rows in set (0.09 sec)
```

5. Please list the numbers of each states' orders which are not delivered.

```
mysql> source 5.sql
```

state	cnt
SP	1245
BA	124
MG	281
SC	91
RJ	502
RS	121
CE	57
MA	30
PR	122
PE	59
DF	60
PB	19
PI	19
MT	21
GO	63
SE	15
ES	38
PA	29
MS	14
AM	3
AP	1
RN	11
TO	6
RO	10
AL	16
RR	5
AC	1

```
27 rows in set (0.04 sec)
```

6. There's a missing order in payment.csv, list its order_id.

```
mysql> source 6.sql
```

order_id
bfbdf9bdef84302105ad712db648a6c

```
1 row in set (0.79 sec)
```

7. Please list the order_id which cost the most days to deliver (the date of product arrival - the date the product starts to deliver) in each state. You must sort states in decreasing order and have **"ORDER BY"** syntax in usage of SQL.

```
mysql> source 7.sql
+-----+-----+
| customer_state | order_id |
+-----+-----+
| TO             | f5691c2b1ca263490374d13d020bd950 |
| SP             | 2fe324feb907e3ea3f2aa9650869fa5 |
| SE             | 285ab9426d6982034523a855f55a885e |
| SC             | a622cb9542d08e3004c1a9a33a7cd859 |
| RS             | dfe5f68118c2576143240b8d78e5940a |
| RR             | 2fa29503f2ebd9f53deba187160f3202 |
| RO             | 8489131c7dcc42a6d6d7df1be6802643 |
| RN             | 3566eabb132f8d64741ae7b921bbd10e |
| RJ             | 1b3190b2dfa9d789e1f14c05b647a14a |
| PR             | d1f789756b59a9fb40b3becadeac90e4 |
| PI             | 2fb597c2f772eca01b1f5c561bf6cc7b |
| PE             | 031e7d4e559a1bf08e71a419aa998d0a |
| PB             | 633f4fc75774571fccae5f584dfce5a7b |
| PA             | 440d0d17af552815d15a9e41abe49359 |
| MT             | 87f795f6abd4bbe786854091a9c6e651 |
| MS             | fc3830b4f000e216dc581df9a0e873e2 |
| MG             | c27815f7e3dd0b926b58552628481575 |
| MA             | 4fbc8d6f2f4db3e789d5a876fa349b56 |
| GO             | 2ba1366baecad3c3536f27546d129017 |
| ES             | ca07593549f1816d26a572e06dc1eab6 |
| DF             | 27167c20a18a136683606cb93c0168a9 |
| CE             | df6d8b7768a047c2981bae0a24afbb01 |
| BA             | 525e11b26fdb7f41471d289897d0f6da |
| BA             | a452fba32eab28a4a62af18eed010c0b |
| AP             | 437222e3fd1b07396f1d9ba8c15fba59 |
| AM             | 28f12a3e5117f5e70bbdb45be9188639 |
| AL             | d573a47c29466bb2e4e46aef0ff04f23 |
| AC             | 0b7cb1c7ea62d95ba39702f4ae850108 |
+-----+-----+
28 rows in set (0.50 sec)
```

8. Please list the number of orders paid by debit card and purchase in 2017 in state 'RJ' .


```
mysql> source 8.sql
+-----+
| cnt |
+-----+
| 57 |
+-----+
1 row in set (0.03 sec)
```

9. Please list the top three seller_ids who sold the most items priced over 150 and the number of these items they sold (price over 150). Only consider sellers who sold more than 100 items. (Hint: Use **WITH**)

```
mysql> source 9.sql
+-----+
| seller_id |
+-----+
| 4869f7a5dfa277a7dca6462dcf3b52b2 |
| 7c67e1448b00f6e969d365cea6b010ab |
| fa1c13f2614d7b5c4749cbc52fecda94 |
+-----+
3 rows in set (0.12 sec)
```

10. Please list the payment_type that appears more than 100 times and the number they appear, **excluding the top two** most frequent payment methods. Order by the number the payment_type appears.

```
mysql> source 10.sql
+-----+-----+
| payment_type | cnt |
+-----+-----+
| voucher      | 2739 |
| debit_card   | 1527 |
+-----+-----+
2 rows in set (0.06 sec)
```